Writer Identification and Verification from Handwritten Digits with Neural Networks

STA5073Z Data Science for Industry 2026 — Assignment 2

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1 Introduction

[Write in your own words: the problem (who wrote a digit, and are two digits by the same person), why it matters (e.g. forensic document examination, signature verifi-

cation (Bromley et al. 1993)), the two approaches compared (a CNN classifier and a Siamese network), and a one-paragraph preview of the main findings.]

All code is in the accompanying bundle: `src/` holds the modules, `run_experiments.py` trains every model and saves results to `results/`, and this report reads those results. The analysis was done in Python with Keras 3 on TensorFlow.

2 Data and exploratory analysis (Task 1)

Table 1: Structure of the dataset.

Check	Value
Images	25,000
Image size	28 x 28
Pixel range (raw)	0 to 255
Missing values	0
Writers / digits / sessions	20 / 5 / 5
Images per writer x digit x session	50
Blank pixels (%)	84.7

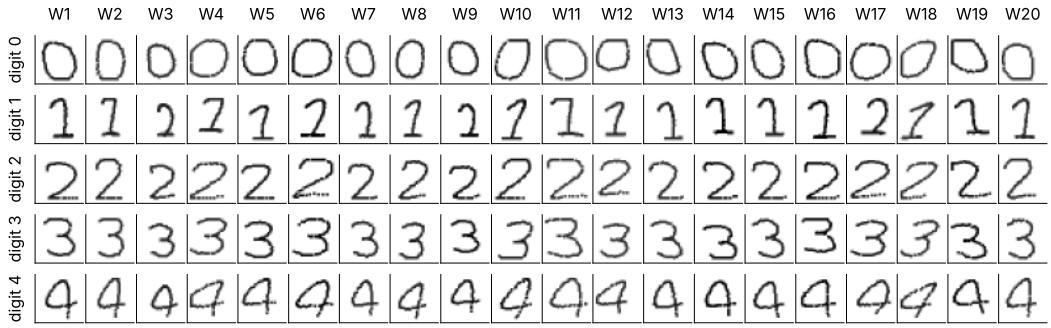


Figure 1: Digits 0–4 (rows) for each of the 20 writers (columns); session 1, replicate 1.



Figure 2: Writer 3: four replicates in each of the five sessions.

[Describe the visible sources of variation: between writers (size, slant, aspect ratio, stroke weight, letter forms such as the hooked 1), between sessions (e.g. writer 3's digit 1 in session 1 vs later sessions), and within a session (replicates nearly identical).]

Table 2: Share of total pixel variance explained by each grouping (between-group sum of squares over total sum of squares).

Grouping	Groups	Share of pixel variance
Digit	5	0.133
Writer	20	0.111
Session	5	0.006
Writer x digit	100	0.475
Writer x digit x session	500	0.914

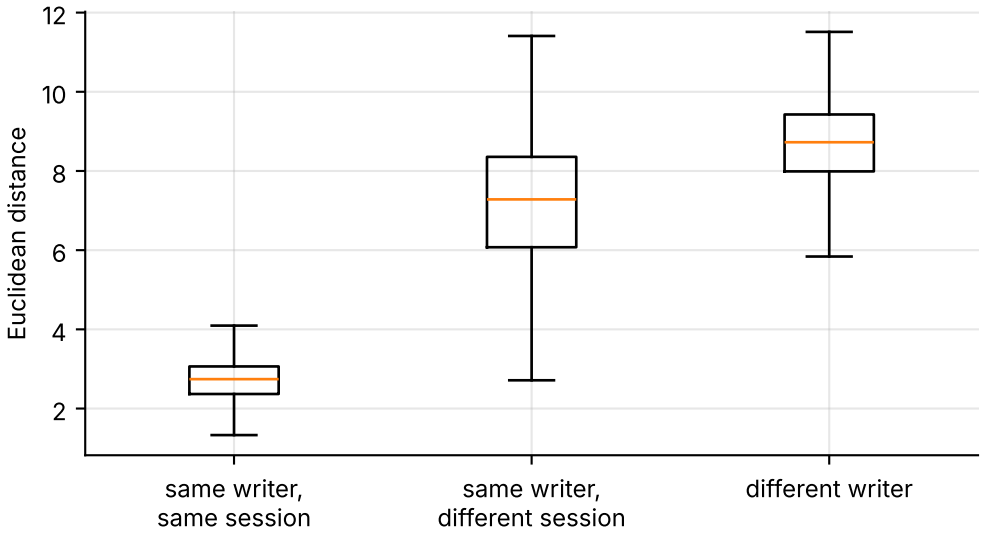


Figure 3: Pixel distance between two images of the same digit, by type of pair (5,000 random pairs of each type).

Median distances are 2.7 (same writer, same session), 7.3 (same writer, different session) and 8.7 (different writers). Once writer and digit are known, adding the session raises the explained pixel variance from 0.475 to 0.914.

[Interpret: what these numbers say about the size of the session effect relative to writer differences, and why the near-duplicate replicates matter for evaluation (link forward to the evaluation design).]

Limitations of the simulated data. [Your list: e.g. only digits 0–4 and 20 writers; perfectly balanced; clean, centred, noise-free images with stamped strokes; session effects that look like simple parametric shifts; unknown generator settings, so models may learn the simulator's rules rather than handwriting traits; effective sample size closer to the 500 writer × digit × session cells than to 25,000 images.]

3 Evaluation design and leakage (Task 4)

Table 3: Train, validation and test strategy.

Model	Question	Train	Validation (early stopping, threshold, tuning)	Test	Generalisation measured
CNN	Which of the 20 writers?	Sessions 1–3, all writers	Session 4	Session 5	Known writers, new recording occasion
Siamese	Same writer or not?	12 writers, all sessions	4 other writers	4 unseen writers (5 rotating folds)	New writers
CNN vs Siamese (Task 6)	Both	Sessions 1–3	Session 4	Session 5	Known writers, new occasion

Table 4: Effect of the split on measured performance.

Model	Evaluation	Metric	Value
CNN	random image split (60/20/20)	accuracy	0.999
CNN	session split (test = session 5), no augmentation	accuracy	0.564
Siamese	unseen writers, same-session pairs	AUC	0.997
Siamese	unseen writers, different-session pairs	AUC	0.876

[Justify the choices in Table 3, explain what each test represents in practice, and discuss leakage and optimism. Points to cover: random splits put near-duplicate replicates in train and test (Table 4); writers cannot be held out for closed-set identification; validation data are reused for early stopping, thresholds and tuning; the decision to add augmentation was made after seeing session-5 accuracy (mild test reuse); only 4 test writers per fold; simulated data.]

4 Writer identification with a CNN (Task 2)

Prediction task. The input is a single 28 × 28 image, scaled to $[0, 1]$; the response is the writer ID (20 classes). The digit is not given to the model. The CNN follows the Lecture 4 pattern (Table 5), is trained with categorical cross-entropy and Adam (batch size 128), and stops early when validation loss has not improved for 5 epochs. A second version adds random rotation (up to about 15°), zoom and shift (up to 10%) to the training images.

Table 5: CNN architecture (`src/cnn.py`); 235,572 parameters in total.

Layer	Output shape	Parameters
Conv 32 filters 3×3, ReLU	26 × 26 × 32	320
Conv 32 filters 3×3, ReLU	24 × 24 × 32	9,248
Max pool 2×2, dropout 0.25	12 × 12 × 32	0
Conv 64 filters 3×3, ReLU	10 × 10 × 64	18,496
Max pool 2×2, dropout 0.25	5 × 5 × 64	0
Flatten, dense 128 ReLU, dropout 0.5	128	204,928
Dense 20, softmax	20	2,580

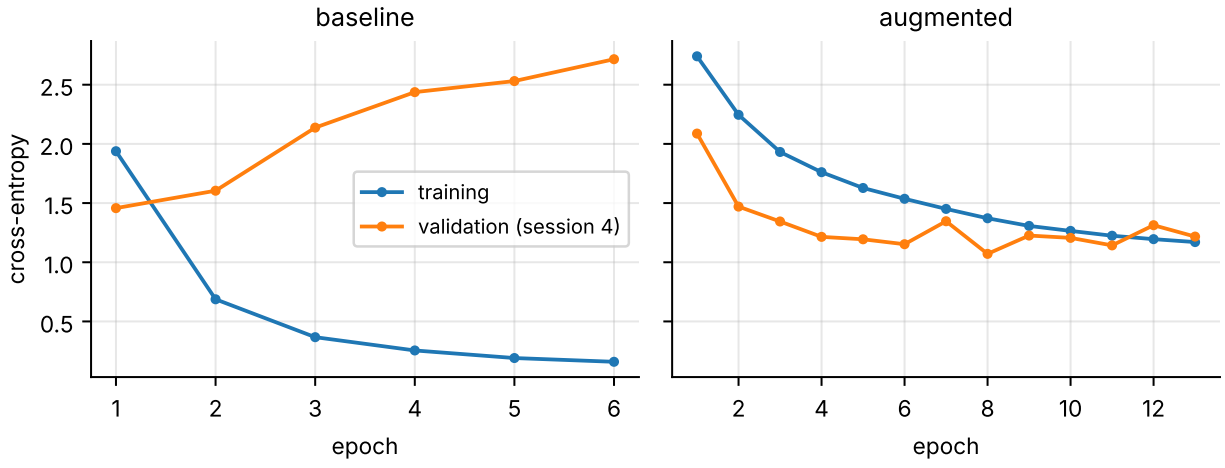


Figure 4: Training and validation loss of the CNN without (left) and with (right) data augmentation.

Table 6: CNN writer identification on the test session (session 5, 5,000 images). Chance accuracy is 0.05.

	Accuracy (top-1)	Top-3 accuracy	Macro F1	Epochs (best + patience)
baseline	0.564	0.855	0.551	6
augmented	0.635	0.905	0.613	13

Table 7: Augmented CNN on session 5, by digit (1,000 images each).

digit	accuracy	top-3 accuracy
0	0.702	0.950
1	0.513	0.868
2	0.673	0.933
3	0.666	0.869
4	0.623	0.905

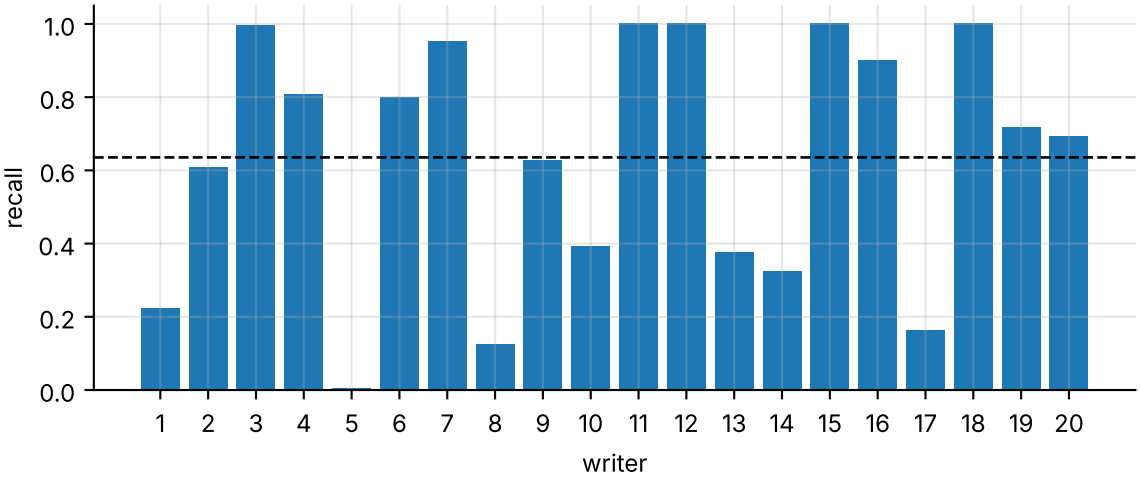


Figure 5: Share of each writer's session-5 images identified correctly by the augmented CNN (dashed: overall accuracy).

Table 8: Accuracy when the CNN's probabilities are averaged over k session-5 images of the same writer.

images per decision	accuracy
1	0.631
2	0.683
3	0.699
5	0.712
10	0.725

images per decision	accuracy

[Report and interpret: overall accuracy 0.635 (vs chance 0.05); overfitting of the baseline and the effect of augmentation (Figure 4); differences by digit and by writer; what accuracy, top-3 accuracy, macro F1 and the multi-image results mean in real-world terms (e.g. a shortlist for a human examiner; more handwriting gives more reliable attribution); the closed-set limitation (an unknown writer is always assigned to one of the 20).]

5 Siamese network for writer matching (Task 3)

Design. A shared embedding CNN (same convolutional layers as Table 5, then dense 128 and a 64-unit output scaled to length 1) maps each image to a vector. Training uses the triplet loss (Schroff et al. 2015), following the Keras example in the course notes (Essam and Valdarrama 2021):

$$L(A, P, N) = \max\big(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha,\ 0\big)\,, \qquad \alpha = 0.5.$$

Two images are declared to be by the same writer when the squared distance between their embeddings is below a threshold chosen on validation writers (`src/siamese.py`).

Table 9: Construction of training triplets; 15,000 new random triplets are drawn every epoch.

Image	Writer	Digit	Session
Anchor A	random	random	random
Positive P	same as A	same as A	different from A
Negative N	different from A	same as A	random

[Explain why the digit is held fixed in every triplet and why positives come from a different session (link to the EDA).]

As a comparison, a second Siamese network with **contrastive loss** (Hadsell et al. 2006) was implemented by Lehlonono Nkejane (originally in R; ported to Python in `src/contrastive.py`): a smaller encoder (16-dimensional embedding), one fixed set of 6,000 training pairs (same digit, different sessions, half same-writer), 10 epochs. Both models are evaluated on the same test pairs.

Table 10: Matching performance on unseen writers: different-session, same-digit test pairs (6,000 per fold, half same-writer). AUC by fold; error rates use the threshold chosen on the validation writers.

fold	Triplet Siamese	Contrastive Siamese	Raw pixels
0	0.805	0.765	0.681
1	0.932	0.915	0.807
2	0.778	0.836	0.769
3	0.946	0.949	0.774
4	0.919	0.838	0.777
mean	0.876	0.860	0.762
sd	0.078	0.072	0.047

Table 11: Error rates on unseen writers (mean over the five folds).

	EER	false accept	false reject	accuracy
Triplet Siamese	0.199	0.254	0.164	0.791
Contrastive Siamese	0.222	0.224	0.231	0.772

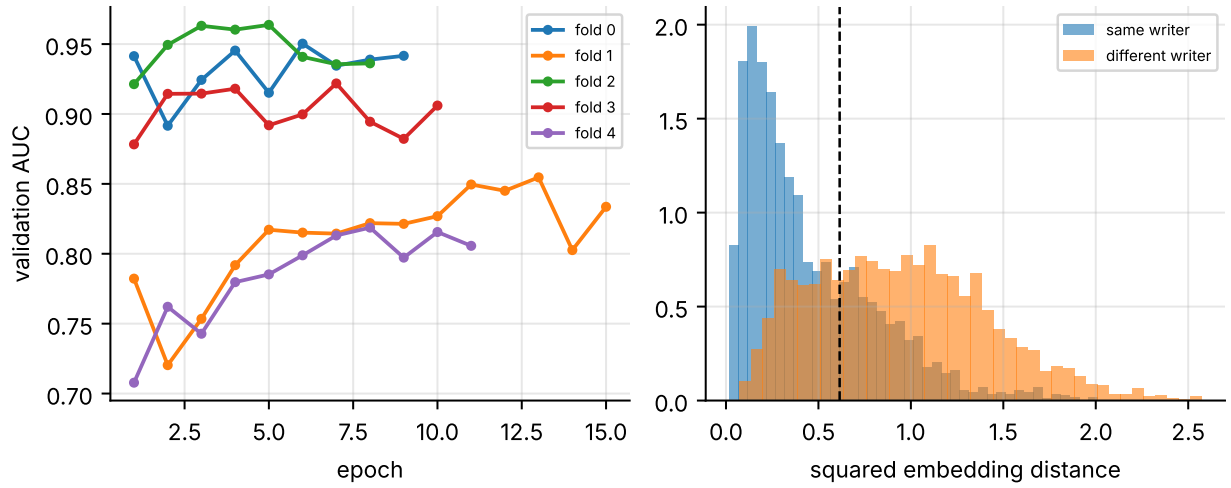


Figure 6: Left: validation AUC (unseen writers) by epoch for each fold. Right: embedding distances of same-writer and different-writer test pairs in fold 0, with the threshold chosen on the validation writers.

Table 12: Triplet Siamese AUC on unseen writers by digit (mean and standard deviation over the five folds).

digit	mean	std
0	0.879	0.088

digit	mean	std
1	0.854	0.064
2	0.897	0.075
3	0.884	0.088
4	0.864	0.125

[Explain how matching performance is evaluated (AUC, equal error rate, false accept and false reject rates at a validation-chosen threshold) and what each means in practice. Interpret: performance on unseen writers vs raw pixels; triplet vs contrastive (similar AUC, contrastive much faster to train); the spread across folds; by-digit differences; same-session vs different-session vs different-digit pairs (Table 4 and appendix Table 19).]

6 Hyperparameter sensitivity (Task 5)

The triplet margin (0.2, 0.5, 1.0) and the embedding size (16, 64) were varied for the Siamese network, trained on sessions 1–3 of all writers and scored on session 4 only. The default (0.5, 64) was repeated with three seeds to measure run-to-run variation.

Table 13: Validation results (session 4) for each setting. AUC: same-writer vs different-writer pairs of a session-4 and a session 1–3 image; identification: nearest-prototype accuracy over 20 writers.

margin	embedding size	seed	epochs	val AUC	val EER	val identification	val top-3
0.200	16	0	6	0.939	0.135	0.584	0.798
0.200	64	0	6	0.936	0.143	0.601	0.801
0.500	16	0	12	0.943	0.136	0.579	0.822
0.500	64	0	15	0.948	0.130	0.608	0.847
1.000	16	0	11	0.934	0.144	0.556	0.817
1.000	64	0	12	0.939	0.138	0.552	0.801
0.500	64	1	15	0.949	0.127	0.627	0.805
0.500	64	2	7	0.943	0.136	0.587	0.850

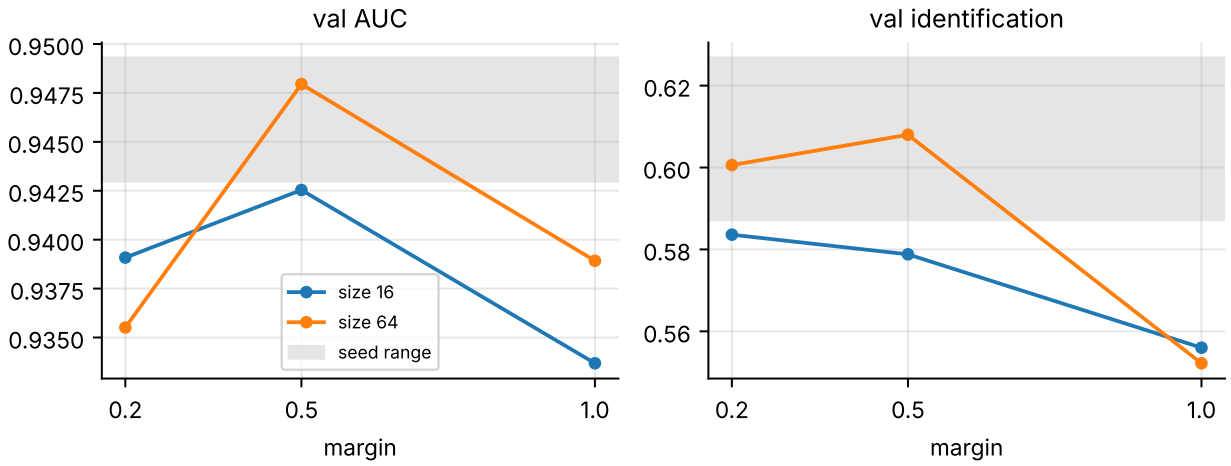


Figure 7: Sensitivity of validation AUC (left) and identification accuracy (right) to the triplet margin and embedding size. Grey band: range over three seeds of the default setting.

Across the grid, validation AUC ranges from 0.934 to 0.948; the three seeds of the default setting range from 0.943 to 0.949. Identification accuracy ranges from 0.552 to 0.608 across the grid.

[Interpret: how sensitive each measure is relative to seed noise; which setting was chosen and why the choice is partly chance; limits of the search.]

7 Comparison of the CNN and the Siamese network (Task 6)

Both models are compared on the session split (test = session 5). The Siamese network identifies writers by the nearest *prototype*, the average embedding of each writer × digit in sessions 1–3 (Snell et al. 2017; Utkin et al. 2019); the CNN's 128-unit feature layer is also used in the same way. For verification, a session-5 image is compared with a session 1–3 image of the same digit.

Table 14: Writer identification on session 5 (20 writers).

model	top-1 accuracy	top-3 accuracy
CNN (softmax)	0.635	0.905
Siamese (nearest prototype)	0.663	0.880
CNN features (nearest prototype)	0.707	0.928

Table 15: Verification on session 5: 20,000 pairs of a session-5 and a session 1–3 image of the same digit; threshold chosen on session 4.

model	AUC	EER	false accept	false reject	accuracy
Siamese	0.949	0.120	0.130	0.107	0.881
CNN features	0.932	0.149	0.148	0.150	0.851
raw pixels	0.772	0.278	0.291	0.269	0.720

Table 16: Writer information by digit on session 5, with the Siamese AUC on unseen writers (Task 3) for reference. Brackets: rank of the digit in each column (1 = best).

digit	CNN identification	Siamese identification	Siamese verification AUC	CNN-feature verification AUC	Siamese AUC, unseen writers
0	0.702 (1)	0.684 (2)	0.951 (3)	0.935 (3)	0.879 (3)
1	0.513 (5)	0.63 (4)	0.939 (5)	0.915 (5)	0.854 (5)
2	0.673 (2)	0.649 (3)	0.948 (4)	0.942 (2)	0.897 (1)
3	0.666 (3)	0.601 (5)	0.958 (1)	0.934 (4)	0.884 (2)
4	0.623 (4)	0.75 (1)	0.953 (2)	0.947 (1)	0.864 (4)

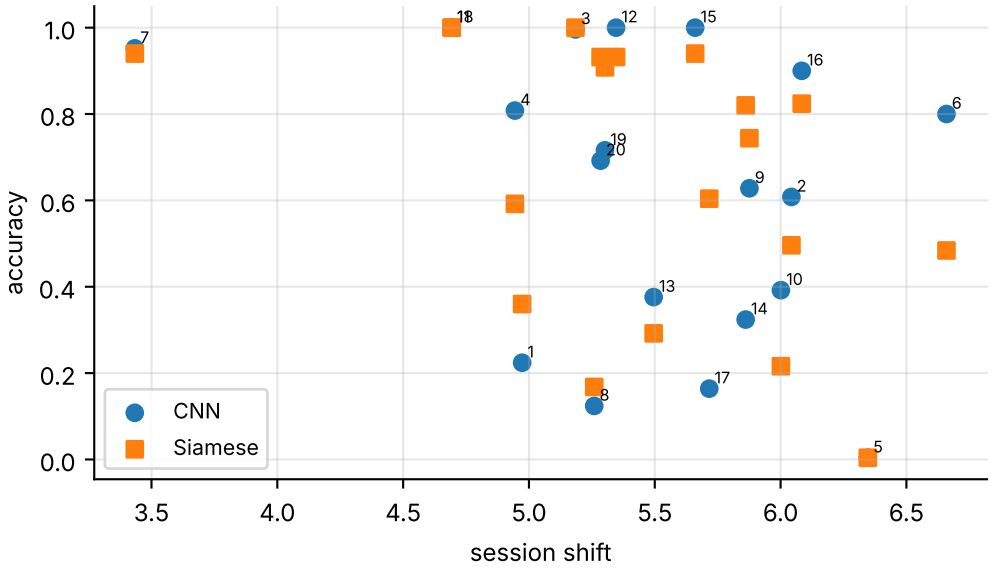


Figure 8: Per-writer session-5 identification accuracy against how much the writer's digits changed between sessions 1–3 and session 5 (pixel distance between average images, averaged over digits).

Spearman correlation between session shift and accuracy over the 20 writers: -0.39 for the CNN ($p = 0.086$) and -0.52 for the Siamese network ($p = 0.019$). Identification

accuracy on session 4 / session 5 is 0.592 / 0.635 for the CNN and 0.627 / 0.663 for the Siamese network.

[Discuss critically: which model suits which task (closed-set identification vs open-set verification; enrolling a new writer without retraining); whether CNN features transfer to verification; which digits carry more writer information, how consistent the ranking is (Table 16) and how large differences are relative to noise (about ±0.015 for 1,000 images); how session changes affect performance (same- vs different-session pairs, session 4 vs 5, per-writer shift in Figure 8, writers that fail completely). Note that the chosen Siamese model is the best of three seeds on validation, which slightly favours it.]

8 Conclusion

[Your conclusions and recommendations, including what would be needed for real handwriting.]

9 Collaboration (Task 7)

Table 17: Commits per GitHub author in the project repository (merge commits are pull requests merged by that author).

author	commits	PR merges
Lungelo-DS	4	3
Lehlonono Nkejane	3	1
LungeloPhewa	2	1
lungelo	2	0
IshmaelNgobeni	1	0

[Describe the workflow: one branch per piece of work, pull requests reviewed by another member before merging, members working on the same files (e.g. the merged EDA notebook combining two members' work); link to the repository and the GitHub Pages site.]

Plagiarism declaration

1. We know that plagiarism is wrong. Plagiarism is to use another's work and pretend that it is one's own.
2. We have used the author–date convention for citation and referencing. Each contribution to, and quotation in, this report from the work(s) of other people has been attributed, and has been cited and referenced.

3. This report is our own work.
4. We have not allowed, and will not allow, anyone to copy our work with the intention of passing it off as his or her own work.
5. We acknowledge that copying someone else's assignment or essay, or part of it, is wrong, and declare that this is our own work.

Signed: Lungelo Phewa _____ Lehlonono Nkejane _____ Ishmael Ngobeni _____ Date: _____

AI use statement

Table 18: AI use, by task.

Task	(a) Tool	(b) What it was used for	(c) How outputs were checked	(d) How much was used
1 EDA	Claude (Anthropic)	Python data loader and EDA code; merging two members' EDA work; checking the R statistics in Python	[to complete]	[to complete]
2 CNN	Claude (Anthropic)	CNN code, augmentation, evaluation code	[to complete]	[to complete]
3 Siamese	Claude (Anthropic)	Triplet-loss Siamese code, pair construction, metrics; Python port of the contrastive model	[to complete]	[to complete]
4 Splits	Claude (Anthropic)	Writer-fold design, leakage comparisons	[to complete]	[to complete]
5 Tuning	Claude (Anthropic)	Grid search and seed-repeat code	[to complete]	[to complete]

Task	(a) Tool	(b) What it was used for	(c) How outputs were checked	(d) How much was used
6 Comparison	Claude (Anthropic)	Prototype identification, comparison code	[to complete]	[to complete]
7 Git	Claude (Anthropic)	Step-by-step Git/GitHub commands	[to complete]	[to complete]
8 Report	Claude (Anthropic)	Quarto structure, figures and tables; no report prose	[to complete]	[to complete]
9 Website	[to complete]	[to complete]	[to complete]	[to complete]

References

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Essam, Hazem, and Santiago L. Valdarrama. 2021. *Image Similarity Estimation Using a Siamese Network with a Triplet Loss*. Keras code example. https://keras.io/examples/vision/siamese_network/.

Hadsell, Raia, Sumit Chopra, and Yann LeCun. 2006. "Dimensionality Reduction by Learning an Invariant Mapping." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2: 1735–42.

Schroff, Florian, Dmitry Kalenichenko, and James Philbin. 2015. "FaceNet: A Unified Embedding for Face Recognition and Clustering." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 815–23.

Snell, Jake, Kevin Swersky, and Richard Zemel. 2017. "Prototypical Networks for Few-Shot Learning." *Advances in Neural Information Processing Systems 30*.

Utkin, Lev V., Maxim S. Kovalev, and Ernest M. Kasimov. 2019. *An Explanation Method for Siamese Neural Networks*. https://arxiv.org/abs/1911.07702.

10 Appendix

Table 19: Triplet Siamese on unseen writers by type of test pair (mean over five folds).

pairs	AUC	EER	accuracy
same_session	0.997	0.023	0.862
diff_session	0.876	0.199	0.791
diff_digit	0.752	0.312	0.601

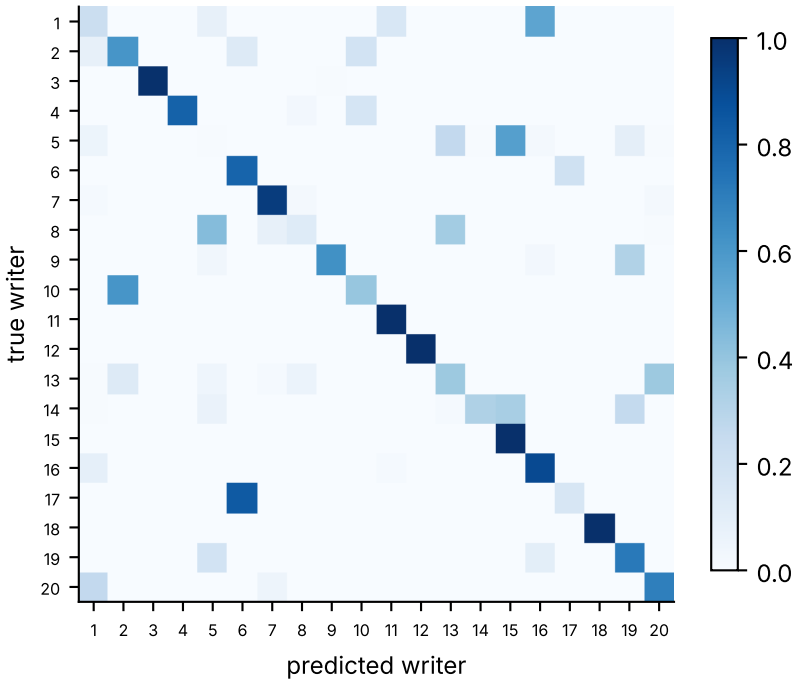


Figure 9: Confusion matrix of the augmented CNN on session 5 (rows: true writer, normalised to sum to 1).

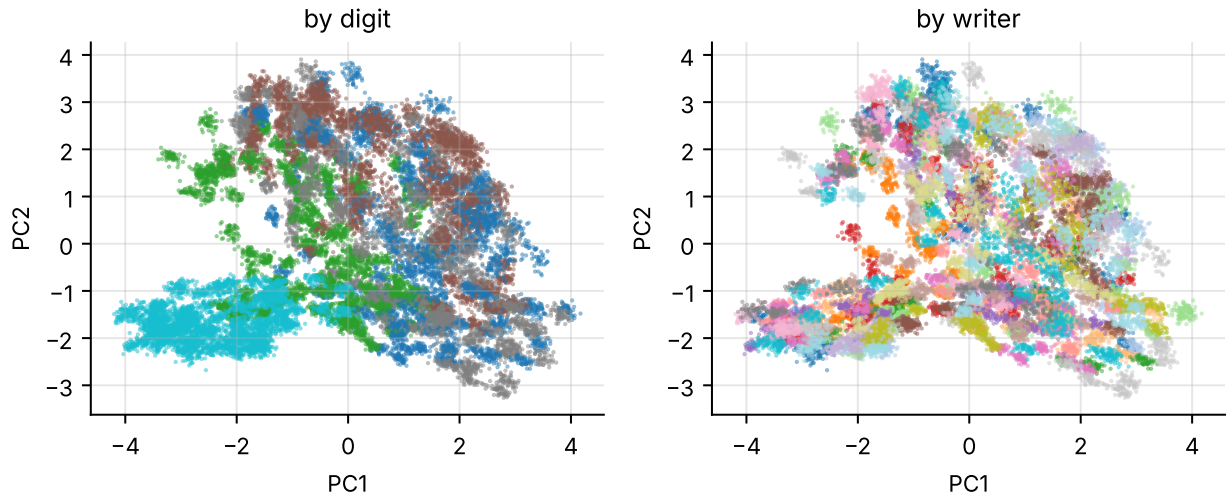


Figure 10: First two principal components of the images, coloured by digit (left) and by writer (right).

The first two components explain 14% of the pixel variance.